Forestry Backpack FEA Report

Purpose

Finite element analysis (FEA) was performed to verify the structural integrity and dynamic performance of the redesigned LiDAR backpack assembly. Static analyses were conducted to evaluate the strength and stiffness of the primary load-bearing components under the project sponsor's specified 5 g loading condition. Modal analyses were performed to identify the assembly's natural frequencies and assess the potential for resonance with the operating frequencies of the rotating LiDAR system.

Assumptions

The following assumptions were made during the analysis:

- Fasteners were assumed to possess greater strength than the surrounding structural components and were therefore modeled using bonded connections rather than explicit bolt representations.
- The backpack mounting interface was represented by fixed boundary conditions applied to the mounting pad locations, approximating attachment to the existing backpack frame.
- Non-structural electronic components (motor controller, Jetson computer, printed circuit boards, cameras, and slip ring) were represented by their equivalent masses to accurately capture inertial loading while minimizing computational complexity.
- The carbon fiber backplate was modeled as an equivalent isotropic material. This simplification was adopted to improve solver stability while providing a reasonable approximation of the global structural response.
- Linear elastic material behavior was assumed for all components.
- Static loading was represented by a uniform 5 g inertial acceleration as specified by the project sponsor.
- Sheet metal components were modeled using both shells and solid geometry depending on material thickness and appropriate simplifications.

Mesh

A curvature-based mesh with automatic local refinement was used for all analyses. This meshing approach provides improved resolution around holes, fillets, and other geometric transitions while maintaining a manageable element count. Because the objective was to evaluate global structural behavior and identify critical stress locations, this mesh was considered sufficient without additional local refinement.

Results

Static Structural Analysis

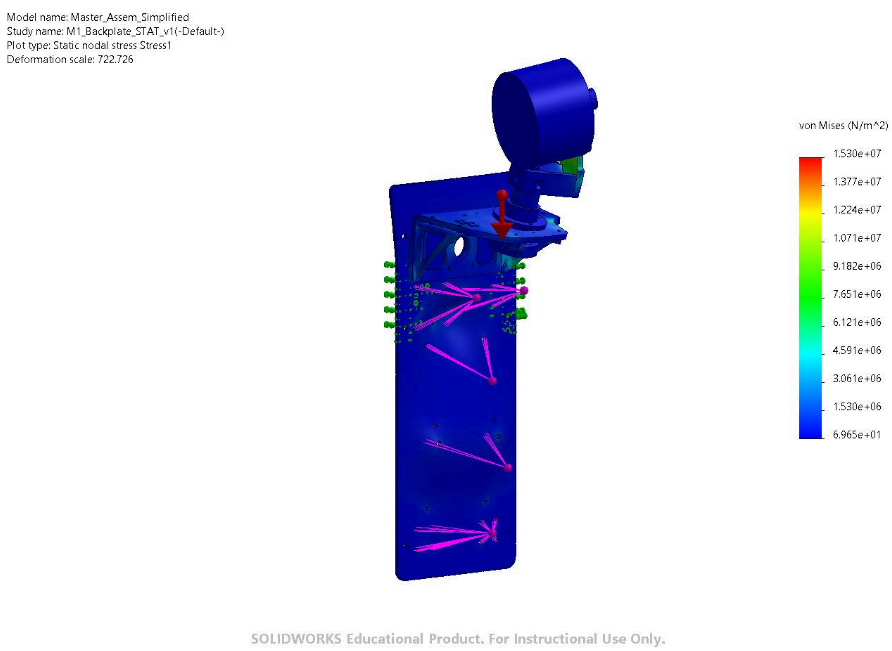

Figure 1. Full Assembly Static Loading

LiDAR Mounting Bracket

Table 1. LiDAR Mounting Bracket Static Results

Metric	Result
Maximum Displacement	0.044 mm
Maximum Von Mises Stress	13.42 MPa
Factor of Safety	20.62

Impact:
The LiDAR mounting bracket remained within acceptable stress limits under the prescribed shock loading condition. Maximum displacement at the LiDAR mounting interface was limited to 0.044 mm, maintaining the positional stability required for LiDAR data collection.

Motor Bracket Assembly

Table 2. Motor Bracket Assembly Static Results

Metric	Result
Maximum displacement	0.048 mm
Maximum Von Mises stress	13.00 MPa
Factor of Safety	4.95

Impact:
The motor bracket assembly maintained structural integrity under the 5 g loading condition. The analysis confirmed that the bracket geometry and mounting configuration provide sufficient strength to support the rotating LiDAR assembly.

Full Assembly

Table 3. Full Assembly Static Results

Metric	Result
Maximum displacement	0.085 mm
Maximum Von Mises stress	15.30 MPa
Factor of Safety	4.27

Impact:
The carbon fiber backplate demonstrated sufficient stiffness and strength to support the redesigned LiDAR system. The maximum displacement remained within acceptable limits, indicating that the plate provides an appropriate structural foundation for the attached components.

Modal Analysis

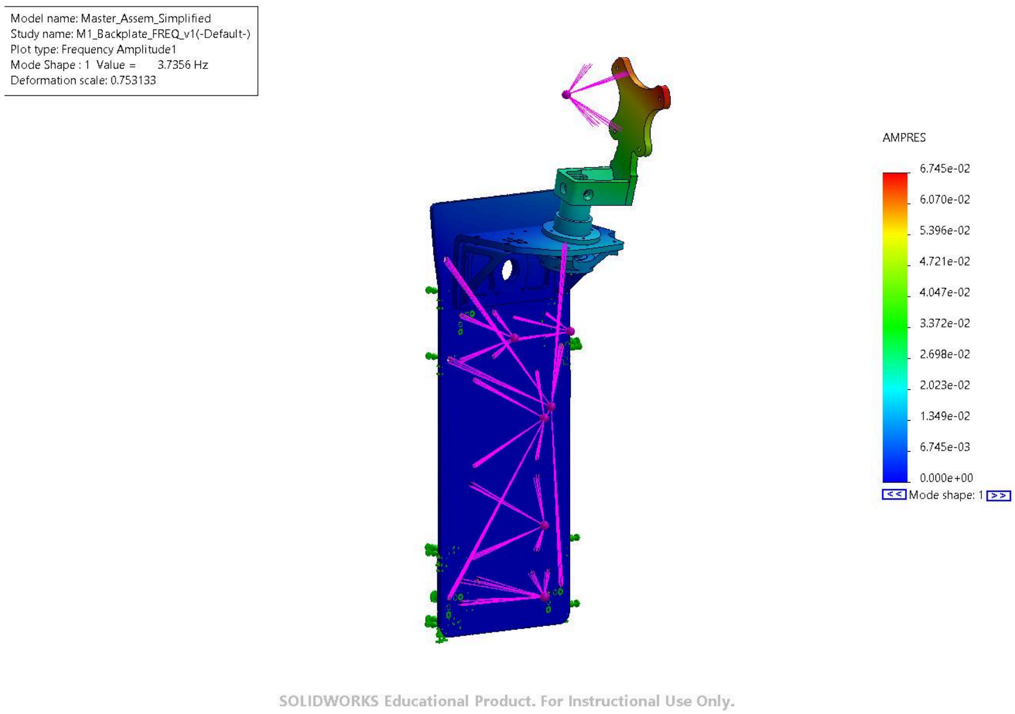

Figure 2. Full Assembly Modal Analysis

The natural frequencies of the assembled structure were evaluated to determine whether operational excitation frequencies could produce resonance. The first several modal frequencies are summarized below.

Table 4. Modal Frequencies

Mode	Natural Frequency (Hz)
1	3.7699
2	4.7698
3	11.538
4	72.415
5	80.852

The system's primary excitation frequencies are:

Table 5. Excitation Frequencies

Source	Frequencies (Hz)
LiDAR internal rotation	10
Secondary LiDAR axis rotation	7
Motor rotation	35

Impact:
The modal analysis identified the lowest natural frequency within the range of human walking cadence, indicating a potential susceptibility to excitation from backpack motion and walking harmonics. However, the identified structural modes do not coincide with the primary operating frequencies of the LiDAR scanning system or motor, reducing the likelihood of resonance during normal operation. Additional local stiffening features, including gussets on the motor bracket, were evaluated but produced minimal frequency shifts. This occurs because the structural modes are broadly distributed across the entire assembly rather than localized to individual components, indicating that overall dynamic behavior is governed by global geometry and mass distribution rather than local stiffness.